

Emilien Seiler

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Skills

Coding

Python, C, C++,
R, Github

AI/ML

Deep Learning,
Reinforcement Learning,
Transfer Learning,
Computer Vision,
Pytorch, Pytorch 3D,
PINNs, Transformers,
Time Series Analysis,
HPC, Signal Processing

Data Engineering

Pandas, Polars, GCP,
AWS, Dagster, Prefect,
Github Action
Deployments, MLOps

Mathematics

Numerical Methods,
Algorithms, Scientific
Machine Learning

Academic

Academic research,
teaching,
L^AT_EX typesetting and
publishing

Languages

French: Native,
English: Fluent level C1

Soft Skills

Autonomy, teamwork,
adaptability

References

Prof. Jan S. Hesthaven,
EPFL
jan.hesthaven@epfl.ch
Prof. Pavlos Protopapas
Harvard University
pprotopapas@g.harvard.edu
Mattia Tomasoni
Jules-Gonin Eye Hospital
mattia.tomasoni@fa2.ch

Professional Experiences

- 2025 – 2025  **Liom, Data Engineer.** Zürich, Switzerland.
- Development of a cloud-based data platform.
 - Startup environment working on non-invasive glucose monitoring.
- 2023 – 2024  **Harvard University, Research Assistant.** Boston, USA.
- Development and implementation of a new method, to address the "failure mode" of vanilla PINNs in stiff systems.
 - Practical experience in bibliographic research and maintaining scientific and technical updates.
- 2022 – 2023  **Jules-Gonin Eye Hospital, Data Science Intern.** Lausanne, Suisse.
- Deep learning model for predicting eye disease progression.
 - Computer vision for 3D OCT image.
- 2021 – 2022  **EPFL, Research Assistant.** Lausanne, Switzerland.
- Artificial Intelligence Laboratory in Natural Language Processing.
- Reinforcement Actor Critic Algorithm with T5 Transformer.
- Computer Vision Laboratory and Swiss Plasma Center.
- Deep learning predictions for nuclear fusion disruptions.
- 2019 – 2023  **EPFL, Teaching Assistant.** Lausanne, Switzerland.
- Mathematical Analysis, Computer Science, Statistical Probability, Mechanical Physics.

Education

- 2023 – 2024  **Harvard University. Master Thesis.**
- Title: *Physics-Informed Neural Networks for Solving Differential Equations in Stiffness Regimes: A Multi-Head Architecture and Transfer Learning Approach.* Grade 5.75/6
- 2021 – 2024  **M.Sc. Ecole Polytechnique Federale de Lausanne**
- Computational Science and Engineering. GPA 5.42/6
- 2018 – 2021  **B.Sc. Ecole Polytechnique Federale de Lausanne**
- Environmental Sciences and Engineering. GPA 5.2/6

Research Publications

- 1 **E. Seiler**, W. Lei, and P. Protopapas, "Stiff transfer learning for physics-informed neural networks," *arXiv preprint*, 2025.  URL: <https://arxiv.org/abs/2501.17281>.
- 2 **E. Seiler**, L. Delachaux, C. Eandi, *et al.*, "Importance of oct-derived biomarkers for the recurrence of central serous chorioretinopathy using statistics and predictive modelling," *Scientific Reports*, 2024.  URL: <https://www.nature.com/articles/s41598-024-75275-7>.
- 3 S. Mousavi, A. Garjani, A. Elwakil, *et al.*, "Cohort builder: A software pipeline for generating patient cohorts with predetermined baseline characteristics from medical records and raw ophthalmic imaging data," *Medical Informatics Europe Conference*, 2024.  URL: <https://pubmed.ncbi.nlm.nih.gov/39176584/>.